

Database Management System (CSC265)

Chapter-wise Study & Priority Guide

BSc. CSIT – 4th Semester – Tribhuvan University

Based on TU Exam Papers: 2076, 2078, 2079, 2080, 2080(New), 2081, 2082, Model (2077 paper not available at time of analysis)

■ ×1 Low – appeared once	■ ×2 Medium – appeared twice
■ ×3–4 High priority	■ ×5+ MUST PREPARE

★ = extremely high priority — appeared in 5+ of 8 papers, prepare thoroughly with worked examples.

Unit 1 – Database and Database Users (2 Hrs)

DBMS Approach vs Traditional File Processing (characteristics & advantages) ■ ×5 ★

Years: 2076, 2078, 2080, 2080(New), 2082

Q: Differentiate the traditional file processing system from the database approach; explain the characteristics/advantages of the DBMS approach.

■ **What to read:** Smallest unit, one recurring definitional question. Know 4–5 clean advantages of DBMS over flat-file systems (redundancy control, data independence, concurrent access, integrity, security) and you're covered.

Unit 2 – Database System – Concepts and Architecture (3 Hrs)

Schema, Instance, Data Independence & Three-Schema (ANSI/SPARC) Architecture ■ ×8 ★

Years: 2076, 2078, 2079, 2080, 2080(New), 2081, 2082, Model

Q: Explain schema vs instance, define data independence (logical/physical), and describe the three-schema architecture with a diagram.

DBMS Classification & Architecture Types (centralized vs client/server, classification by data model) ■ ×2

Years: 2081, Model

Q: Classify DBMSs based on the data model they support; differentiate centralized from client/server architecture with an example.

■ **What to read:** The single most reliable topic in the entire subject — schema/instance/data independence shows up in every paper, sometimes as the whole long-answer question, sometimes buried in a short question. Master the three-schema diagram (external/conceptual/internal) and the two independence types cold.

Unit 3 – Data Modeling Using the Entity-Relationship Model (6 Hrs)

ER Diagram Construction (full diagram for a given real-world scenario) ■ ×4 ★

Years: 2078, 2081, 2082, Model

Q: Construct a complete ER diagram for a described system (e.g. bookstore, airline booking, course registration, library) with appropriate entities, attributes, keys and relationships.

Entity Types, Entity Sets, Attributes & Keys ■ ×3

Years: 2076, 2079, 2081

Q: Define entity type and entity set; differentiate entities from entity sets with an example.

Specialization, Generalization & Constraints ■ ×3

Years: 2080(New), 2081, Model

Q: Define specialization and generalization; explain disjoint and other constraints on specialization with an example.

Relationship Types, Relationship Sets, Roles & Weak Entity Types ■ ×2

Years: 2078, 2080(New)

Q: Explain relationship types and relationship sets with an example; define weak entity type.

Aggregation ■ ×1

Years: 2079

Q: Explain the concept of aggregation in ER modeling with an example.

■ **What to read:** ER diagram construction is the guaranteed long-answer question — practice drawing full diagrams (with cardinality, keys, weak entities) for 4–5 different scenarios (e-commerce, library, banking, booking systems) so the pattern is automatic. Specialization/generalization is the next most tested sub-concept.

Unit 4 – The Relational Data Model and Relational Database Constraints (3 Hrs)

Relational Constraints (domain, entity integrity, referential integrity) ■ ×5 ★

Years: 2076, 2078, 2081, 2082, Model

Q: Define relational constraints; explain domain constraint, entity integrity and referential integrity with examples.

Relational Model Concepts (domain, attribute, tuple, relation) ■ ×2

Years: 2080, 2080(New)

Q: Define the relational model and the terms domain, attribute, tuple and relation; explain the characteristics of a relation.

Integrity vs Security ■ ×1

Years: 2076

Q: Differentiate between integrity and security in the context of a database, with an example.

■ **What to read:** Referential integrity is the anchor sub-topic here — make sure you can explain it with a concrete primary-key/foreign-key example, not just the definition. This unit is short but its constraints vocabulary underlies nearly every SQL and normalization question elsewhere in the paper.

Unit 5 – The Relational Algebra and Relational Calculus (5 Hrs)

SQL & Relational Algebra Query Writing (from a given schema) ■ ×6 ★

Years: 2078, 2079, 2080, 2080(New), 2081, 2082

Q: Given a relational schema (2–3 tables), write both SQL and Relational Algebra expressions for a set of retrieval, count, update and delete queries.

Basic Relational Algebra Operations (UNION, INTERSECT, DIFFERENCE, CARTESIAN PRODUCT, JOIN, NATURAL JOIN) ■ ×5 ★

Years: 2076, 2078, 2080, 2081, 2082, Model

Q: Explain a set-theory relational algebra operation or JOIN/NATURAL JOIN with a suitable example; differentiate JOIN from NATURAL JOIN.

Tuple Relational Calculus ■ ×2

Years: 2080, 2080(New)

Q: Given a schema, write a Tuple Relational Calculus expression for a described retrieval condition.

■ **What to read:** This is the highest-frequency numerical-style unit in the whole subject — the paired SQL + RA query question is asked in 6 of 8 papers, every single time on a 2–3 table schema. Practice writing both SQL and RA side-by-side for SELECT, JOIN, COUNT/GROUP BY, UPDATE and DELETE queries until it's automatic.

Unit 6 – SQL (8 Hrs)

SQL DDL & Constraints (CREATE TABLE with primary/foreign keys) ■ ×1

Years: 2076

Q: Create tables using SQL with primary key, foreign key and referential integrity constraints for a described scenario.

Assertions & Triggers ■ ×1

Years: 2079

Q: Explain the concepts of Assertions and Triggers in SQL with an example.

■ **What to read:** SQL retrieval/update/delete queries are covered above under Unit 5 since every paper pairs them with Relational Algebra on the same schema — treat that as the real Unit 6 workload. Assertions/Triggers and DDL constraints are lower-frequency standalone questions; know the definitions but don't over-invest.

Unit 7 – Relational Database Design (7 Hrs)

Normalization (1NF, 2NF, 3NF with example) ■ ×6 ★

Years: 2076, 2079, 2080, 2080(New), 2082, Model

Q: Define normalization; explain 1NF, 2NF and 3NF with a suitable example, and discuss why normalization is important in database design.

Functional Dependencies (types, trivial/non-trivial, inference rules) ■ ×5 ★

Years: 2078, 2079, 2080(New), 2082, Model

Q: Define functional dependency; explain different types of functional dependencies, and why inference rules are needed.

ER-to-Relational Mapping & Informal Design Guidelines ■ ×3

Years: 2079, 2080(New), 2081

Q: Explain how entities and relationships from an ER diagram are mapped into relational tables; discuss the informal design guidelines for relational schemas.

Boyce-Codd Normal Form (BCNF) & Fourth Normal Form (4NF) ■ ×1

Years: 2078

Q: Define BCNF with example and explain how it differs from 3NF.

■ **What to read:** Normalization (1NF/2NF/3NF with worked example) is a near-guaranteed long-answer question — practice normalizing 2–3 sample unnormalized tables end-to-end so you can show the decomposition, not just define the forms. Functional dependencies are the theoretical backbone underneath — study them together.

Unit 8 – Introduction to Transaction Processing Concepts and Theory (4 Hrs)

Schedules, Serializability & Conflict-Serializability ■ ×6 ★

Years: 2076, 2078, 2079, 2080(New), 2081, 2082

Q: Define schedule; explain conflict-serializability with an example, and how concurrency control ensures serializability.

ACID Properties & Desirable Properties of Transactions ■ ×5 ★

Years: 2079, 2080, 2081, 2082, Model

Q: Explain the ACID properties of a transaction, or the desirable properties of transactions in general.

Buffer Management ■ ×1

Years: 2079

Q: Explain the concept of Buffer Management in DBMS.

■ **What to read:** Serializability/conflict-serializability is the single most-repeated concept in this unit — be able to test a given schedule for conflict-serializability step by step, not just define it. Pair it with ACID properties since they're often asked as a combined or immediately-following question.

Unit 9 – Concurrency Control Techniques (4 Hrs)

Two-Phase Locking (2PL) & Lock Types ■ ×4 ★

Years: 2076, 2080, 2082, Model

Q: Explain the Two-Phase Locking protocol; describe basic, conservative, strict and rigorous 2PL variants, lock conversion, and differentiate binary lock from shared/exclusive lock.

Concurrency Control Techniques (Timestamp Ordering, Optimistic, Multiversion) ■ ×3

Years: 2078, 2080(New), 2082

Q: Explain timestamp ordering protocol, or optimistic and multiversion concurrency control techniques.

Deadlock (prevention, detection, starvation) ■ ×3

Years: 2078, 2080(New), 2081

Q: Explain deadlock with an example; discuss deadlock prevention protocols, detection, and how starvation relates to deadlock.

Granularity of Data Items ■ ×1

Years: 2076

Q: Explain the concept of granularity of data items and its effect on concurrency control.

■ **What to read:** Two-Phase Locking is the anchor numerical/conceptual topic — know all four variants (basic, conservative, strict, rigorous) and lock conversion, since exams like to ask you to distinguish them. Deadlock and alternative concurrency techniques (timestamp/optimistic/multiversion) round out this unit.

Unit 10 – Database Recovery Techniques (3 Hrs)

Recovery Techniques (deferred update, immediate update, shadow paging, checkpoints) ■ ×7 ★

Years: 2076, 2078, 2080, 2080(New), 2081, 2082, Model

Q: Explain why database recovery is essential and describe one or more recovery techniques (deferred update, immediate update, shadow paging) or the role of checkpoints in recovery.

■ **What to read:** Short unit but the single most consistently-asked topic outside Units 2, 5, 7 and 8 — it appears in nearly every paper, usually as one dedicated question. Know all three recovery techniques (deferred update, immediate update, shadow paging) well enough to explain any one in detail, since the exact one asked varies by year.

Top Priority Summary (Sorted by Frequency)

Topic	Unit	Freq	Years
Schema/Instance/Data Independence/3-Schema Arch.	DB System Concepts & Architecture	×8	2076–82, Model
Database Recovery Techniques	Database Recovery	×7	2076, 78, 80, 80(N), 81, 82, Model
SQL & Relational Algebra Query Writing	Relational Algebra & Calculus	×6	2078, 79, 80, 80(N), 81, 82
Normalization (1NF/2NF/3NF)	Relational Database Design	×6	2076, 79, 80, 80(N), 82, Model
Schedules & Serializability	Transaction Processing	×6	2076, 78, 79, 80(N), 81, 82
Basic RA Operations (Union/Join/Natural Join etc.)	Relational Algebra & Calculus	×5	2076, 78, 80, 81, 82, Model
Relational Constraints (domain/entity/referential)	Relational Data Model	×5	2076, 78, 81, 82, Model
Functional Dependencies	Relational Database Design	×5	2078, 79, 80(N), 82, Model
ACID Properties & Transaction Properties	Transaction Processing	×5	2079, 80, 81, 82, Model
DBMS vs Traditional File Processing	DB and Database Users	×5	2076, 78, 80, 80(N), 82
ER Diagram Construction	Entity-Relationship Model	×4	2078, 81, 82, Model
Two-Phase Locking & Lock Types	Concurrency Control	×4	2076, 80, 82, Model
Entity Types, Sets, Attributes & Keys	Entity-Relationship Model	×3	2076, 79, 81
Specialization/Generalization & Constraints	Entity-Relationship Model	×3	2080(N), 81, Model
ER-to-Relational Mapping / Design Guidelines	Relational Database Design	×3	2079, 80(N), 81
Alt. Concurrency Techniques (Timestamp/Optimistic/MVCC)	Concurrency Control	×3	2078, 80(N), 82
Deadlock (prevention/detection/starvation)	Concurrency Control	×3	2078, 80(N), 81

Final Exam Very High Probability Topics (Never Miss These)

MUST PREPARE (×5+, near-certain): Schema/instance/data independence, Database recovery techniques, SQL & RA query writing, Normalization (1NF/2NF/3NF), Schedules & serializability, Basic RA operations, Relational constraints, Functional dependencies, ACID properties, DBMS vs file processing.

High priority (×3–4): ER diagram construction, Two-phase locking, Entity types/sets, Specialization/generalization, ER-to-relational mapping, Alternative concurrency techniques, Deadlock.

Numerical / worked-example heavy (practice by hand): Writing SQL + RA side-by-side for a given schema, normalizing an unnormalized table through 1NF→3NF, testing a schedule for conflict-serializability, drawing a full ER diagram for a scenario.

Due for a repeat (asked before, quiet recently — good bet): BCNF/4NF, Buffer management, Granularity of data items, Tuple relational calculus, Aggregation, Integrity vs security, Assertions & Triggers.